ECE101L Lab3

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1 RC Integrator

1.1 Introduction

An RC circuit is any circuit that includes at least one capacitor. Capacitors are more complex than resistors, since the values measured in a capacitor depend on the rate of change of the inputs. Commonly, these changes take place due to AC sources, digital signals, and analog signals. Sinusoidal, square, and complex waves. Physically, capacitors are a pair of separated plates that are built to allow static electrical charge to build on them, then release them when later. Because they are separated, they should be able to act like an opening under correct conditions.

Using differential equations, capacitors voltage in simple RC circuits can be modeled with:

$$V_C(t) = V_C(\infty) + [V_C(0) - V_C(\infty)]e^{-\frac{t}{\tau}} \quad , \tau = RC$$

Using a square wave, we will provide a changing voltage to the capacitor, allowing us to observe the properties of capacitors and verify this equation.

I predict that the equation will closely model the capacitor. And That the capacitor, given time, will match the same voltage as if it were an open circuit.

1.2 Experiment

Our circuit, modeled in Fig 1, consists of a $2k\Omega$ resistor and $1nF$ capacitor. The wave form generator has been set to a $10kHz$ square wave ranging from approximately $0V$ to $10V$. Its important to note that the generator is expected to used with leads of a specific resistance which I did not have, so the actual settings on the machine read $0V$ to $5V$. The generator also has a 50Ω internal resistance. The oscilloscope it hooked onto the capacitor.

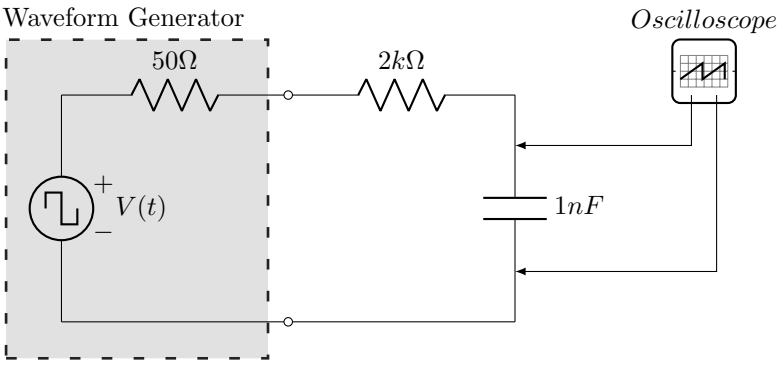


Figure 1: RC Integrator

To start we can measure the capacitors voltage each time one time constant passes. The time constant, τ, is $R_{total} * C$. This is $2.05ns$. We will also be using the equation for $V_C(t)$ to compare results and determine if it matches reality.

For the measurements I actually measured every $2ns$ instead of $2.05ns$ since I was not allowed to adjust any finer. This was reflected in my calculations so can be ruled out as a source of error later. While measuring, it was observed that the highest voltage was $10.211V$ so it was assumed that this was the true output of the waveform generator, since it was known to display the wrong value. Fig 2 displays the measured wave for $V_C(t)$.

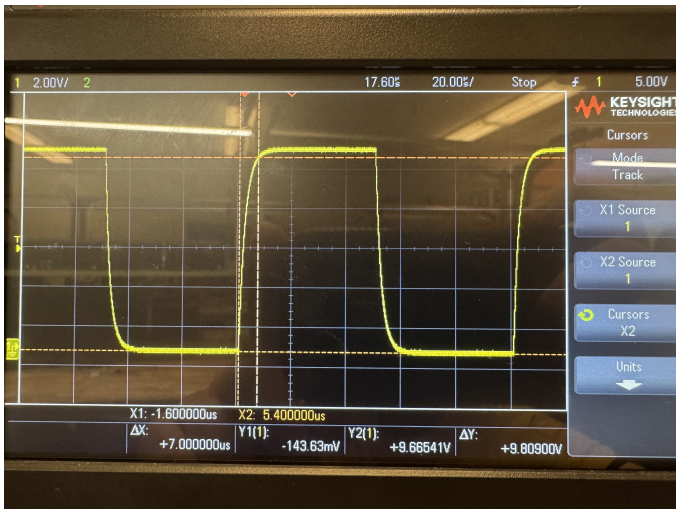


Figure 2: Capacitor Voltage on Oscilloscope

For the equations, we need to make two for when the square wave is high and one when it is low. Since they need a steady state to be applicable they both will assume that the square wave switching states is $t = 0$ and that the wave never switches back. This means for the charging case $V_C(0) = 0$, and $V_C(\infty) = 10.211$. It will be the same for the draining case but reversed. The result is these two equations:

$$\text{Charging: } V_C(t) = 10.211(1 - e^{-\frac{t}{2.05ns}})$$

$$\text{Draining: } V_C(t) = 10.211e^{-\frac{t}{2.05ns}})$$

Table 1 shows the comparisons of all the data as well as, fig 3, and fig 4.

Table 1: Capacitor Voltage by Time Constant

$V(n\tau)$	Charging V_{meas}	Charging V_{pred}	Draining V_{meas}	Draining V_{pred}
$V(\tau)$	$6.030V$	$6.362V$	$3.749V$	$3.849V$
$V(2\tau)$	$8.603V$	$8.760V$	$1.460V$	$1.451V$
$V(3\tau)$	$9.648V$	$9.664V$	$570mV$	$547mV$
$V(4\tau)$	$9.970V$	$10.005V$	$275mV$	$206V$
$V(5\tau)$	10.211	$10.133V$	$4mV$	$0.078mV$

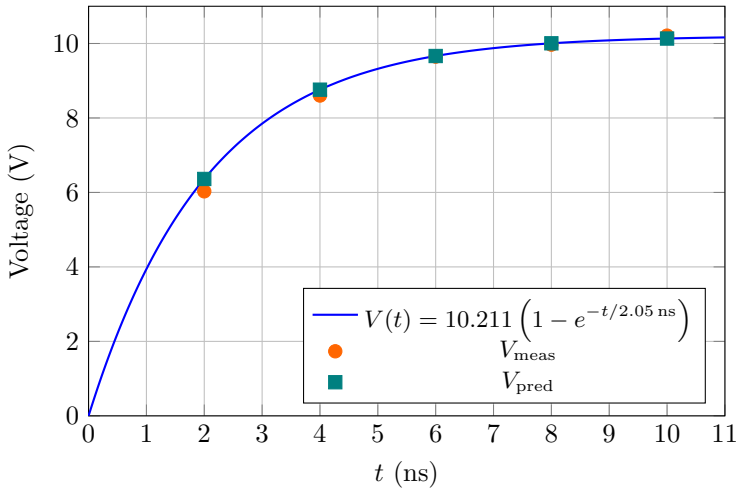


Figure 3: Capacitor Charging Voltage

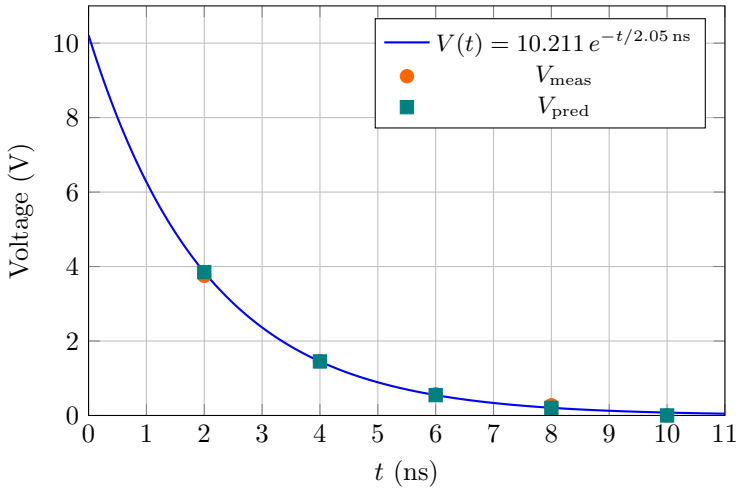


Figure 4: Capacitor Draining Voltage

1.3 Conclusion

For the most part the equations match very well to the measurements. However they are slightly off in some moments. It's also easy to spot that the the equations initially go over the measurements but go under later. The greatest potential cause of this is likely due to how the maximum source voltage was measured, since instead of measuring it directly I used what I saw across the capacitor. This was made the measurement unreliable. In hindsight I should have measured directly off of the waveform generator. But since even with this issue the equations strongly resemble the true values I think it verifies its validity.

2 RC Differentiator

2.1 Introduction

Now we will alter our circuit to measure the voltage across the resistor during the same square wave. The only change made was swapping the positions of the capacitor and resistor, as seen in Fig 5. The waves we see should be able to be related back to what we was in the previous section.

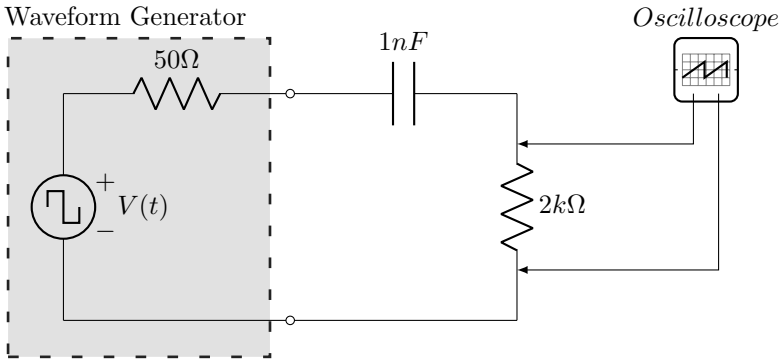


Figure 5: RC Differentiator

We will be repeating the experiment of the last section but in the context of the resistor instead of the capacitor. We will need to derive new equations from the equation of $V_C(t)$.

I think that we will be able to relate the voltage across the resistor and capacitor using KVL, and therefor resistor voltage will have voltage magnitude opposite to the capacitor in the range of the square waves voltage.

2.2 Experiment

Like the last section we will start by measuring the resistors voltage over time as the capacitor drains and charges. We will again use $2ns$ intervals but still accommodate the true time constant in our calculations. Since I am unable to remeasure the max voltage across the capacitor, $10.211V$ will need to be used again. Fig 6 shows the observed wave of $V_R(t)$. Interestingly the wave actually goes negative over some segments. this is because when the capacitor drains it makes current flow in the opposite direction to where it flowed before.

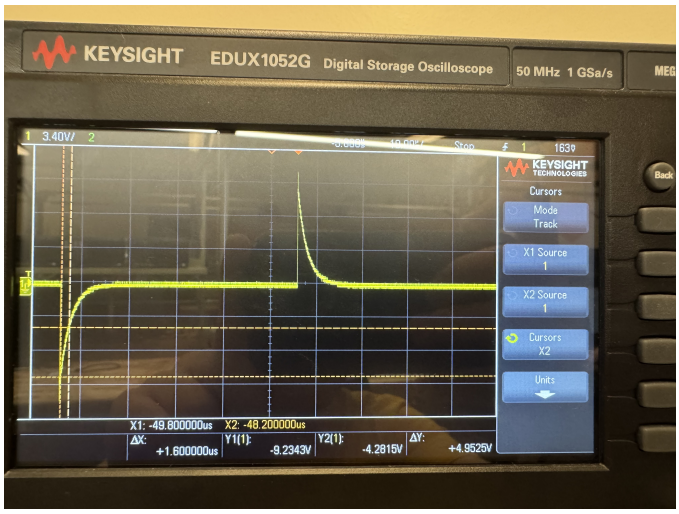


Figure 6: Capacitor Voltage on Oscilloscope

For The equations we will use KVL to solve for the voltage across the resistor. While charging, the voltage across both resistors will be $(V_s - V_C)$ we can then use a voltage divider to find the voltage only across our resistor. For draining, the source can be represented as a short since its off. this means the voltage across both resistors is just V_C and we can use a voltage divider again to solve for just the $2k\Omega$ resistor. This will also be multiplied -1 since the current is flowing in the opposite direction. Interestingly, these equations end up being the same thing except the Draining case is negative:

$$\text{Charging: } V_R(t) = 10.211e^{-\frac{t}{2.05ns}}(\frac{2000}{2000+50})$$

$$\text{Draining: } V_R(t) = -10.211e^{-\frac{t}{2.05ns}}(\frac{2000}{2000+50})$$

Now the data from the predictions and measurements has been compiled in Table 2. The data is also displayed in Fig 7 and Fig 8.

Table 2: Resistor Voltage by Time Constant

$V(n\tau)$	Charging V_{meas}	Charging V_{pred}	Draining V_{meas}	Draining V_{pred}
$V(\tau)$	$3.678V$	$3.755V$	$-3.750V$	$-3.755V$
$V(2\tau)$	$1.379V$	$1.161V$	$-1.451V$	$-1.161V$
$V(3\tau)$	$671mV$	$437mV$	$-677mV$	$-437mV$
$V(4\tau)$	$140mV$	$165mV$	$-213mV$	$-165mV$
$V(5\tau)$	$140mV$	$62mV$	$-213mV$	$-62mV$

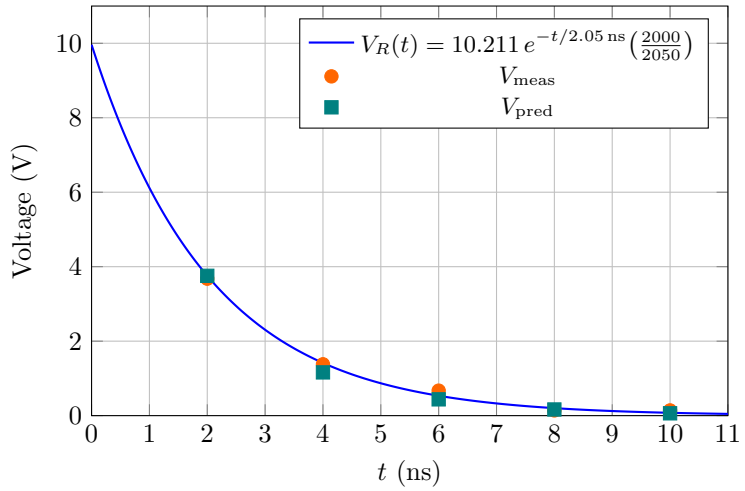


Figure 7: Resistor Voltage During Charging

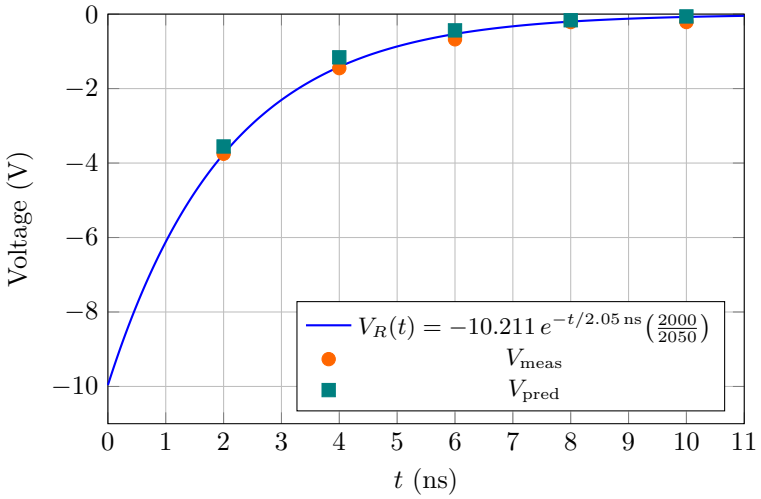


Figure 8: Resistor Voltage During Draining

2.3 Conclusion

Since The measurements line up closely with the predictions, this verifies our understanding of the circuit and further verifies the equation for $V_C(t)$. Again, small divergences suspected to be caused most by assuming the incorrect voltage for when the source is high. Other error sources like electrical noise may also still apply.

Notice how for our charging cases, the resistors voltage magnitude is highest right after the wave turns on, and the capacitor is highest when the wave has already been on for some time. When other people noticed this behavior they named the circuits a high pass and low pass filter respectively, because they pass voltage only when the frequency is high or low. On the resistor, high pass filter, the voltage across it is highest when the frequency is high, like right after the wave goes to its maximum. for the capacitor, low pass filter, the voltage is highest when in a steady or low frequency state, which is why it was highest after the wave stayed at is maximum for some time.

The reason they are called RC integrators and differentiators is because they approximate those mathematical functions. In the resistor, the integrator, the spikes in voltage during switching resemble spikes in the rate of change of the square wave. the flat segments resemble the derivative when rate of change is zero. The capacitor resembles an integral since it would become a triangle wave if it was not given more time to stabilize. Making the square wave higher frequency would remove the ending asymptotic approaches. A triangle wave is the integral of a square wave.

3 RL Circuit

3.1 Introduction

Similar to RC circuits, RL circuits are any circuit with at least one inductor inside. They are dependent to changing values like capacitors but are instead reliant on current, not voltage. They function by using winding coils to make magnetic fields that resist changing current inside of itself. In a steady state they act like a short.

Using differential equations, this equation was made to describe inductor behavior in simple RC circuits.

$$I_L(t) = I_L(\infty) + [I_L(0) - I_L(\infty)]e^{-\frac{t}{\tau}} \quad , \tau = L/R$$

Notice how this is in terms of current instead of voltage, but is otherwise identical to the equation of $V_C(t)$. This will be an issue later since the measurements taken had to be voltage. The purpose of this segment is to observe the behaviors of RL circuits and experimentally derive the time constant, τ, for our circuit.

Since the equations are similar, I think that the current over time will closely resemble the plot for the voltage over time in the RC circuit. I also think the time constant we will find will match up with the real time constant.

3.2 Experiment

Our circuit, pictured in Fig 9, uses the same waveform generator and oscilloscope. We are using a 100Ω resistor and a $10\mu H$ inductor which the oscilloscope is measuring the voltage of. The waveform we measured is pictured in Fig 10. When obtaining the measurements, I chose to record every $10ns$ in the beginning where it changed most rapidly. Later I recorder every $20ns$ where the rate of change was lower. The exception is the peak voltage, since it occurred at an odd time. These measurements are recorded in Table 3.

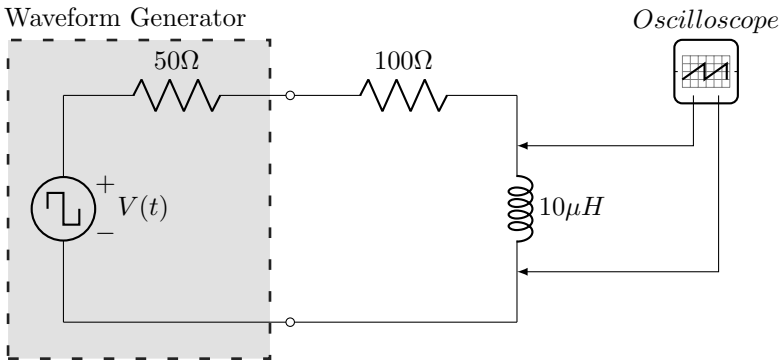


Figure 9: RL Circuit

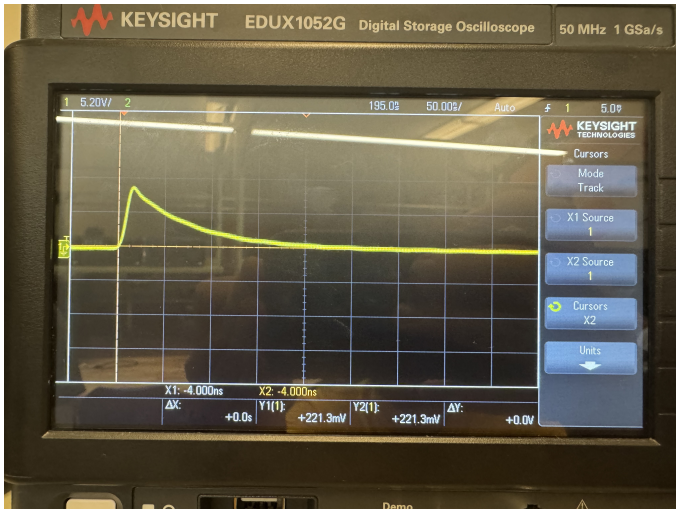


Figure 10: Capacitor Voltage on Oscilloscope

Table 3: RL Circuit Voltage Samples

	1	2	3	4	5	6	7	8
Time (ns)	0	10	18.4(V_{peak})	20	30	40	50	60
Voltage (V)	0	4.3484	9.3592	9.1904	7.3220	6.4754	5.5347	4.8344

	9	10	11	12	13	14	15
Time(ns)	70	80	90	100	130	150	170
Voltage V	4.2700	3.6298	3.1516	2.7309	1.6831	1.4818	1.0716

Here is the problem, we need the time constant to do anything meaningful, but its suppose to be found experimentally. If current was measured instead of voltage we could verify it using the equation for $I_L(t)$ but all the data is in volts. What I decided to do instead was find the best fit curve for our data points. To do this I decided to find a multi exponential using MATLAB. The library responsible for the heavy lifting was "exp". This allowed me to make the function as seen in Fig 11.

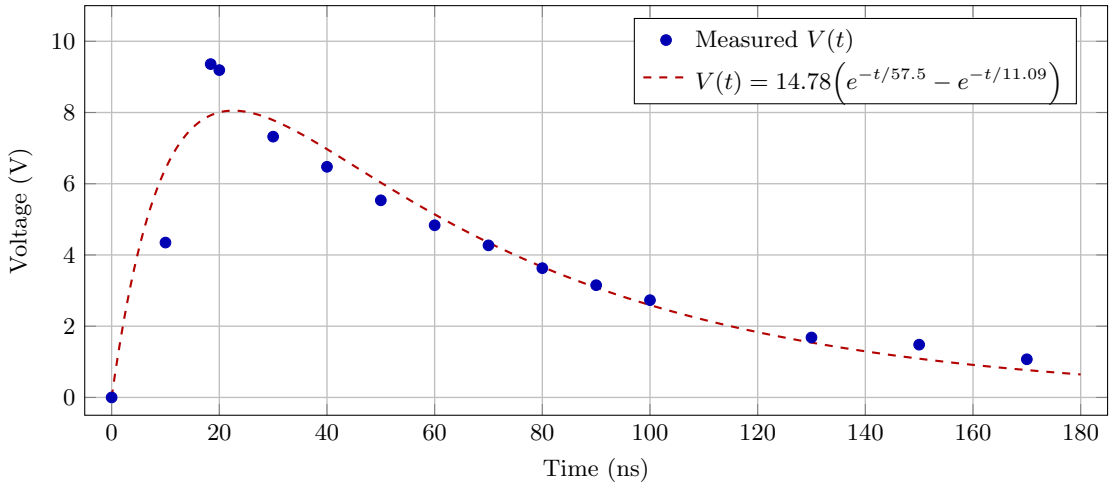


Figure 11: Measured Voltages and Best Fit Equation

Now that we have an approximation, we can use the fact that $V_L = L\frac{dI_L}{dt}$ to make an approximation for current over time. After rearranging, the equation becomes:

$$I_L = \frac{1}{L}\int V_L \, dt$$

The result is plotted in Fig 12 as the equation $I_L(t) = 1.478(-57.5e^{-\frac{x}{57.5}} + 11.09e^{-\frac{x}{11.09}} + 46.41)mA$

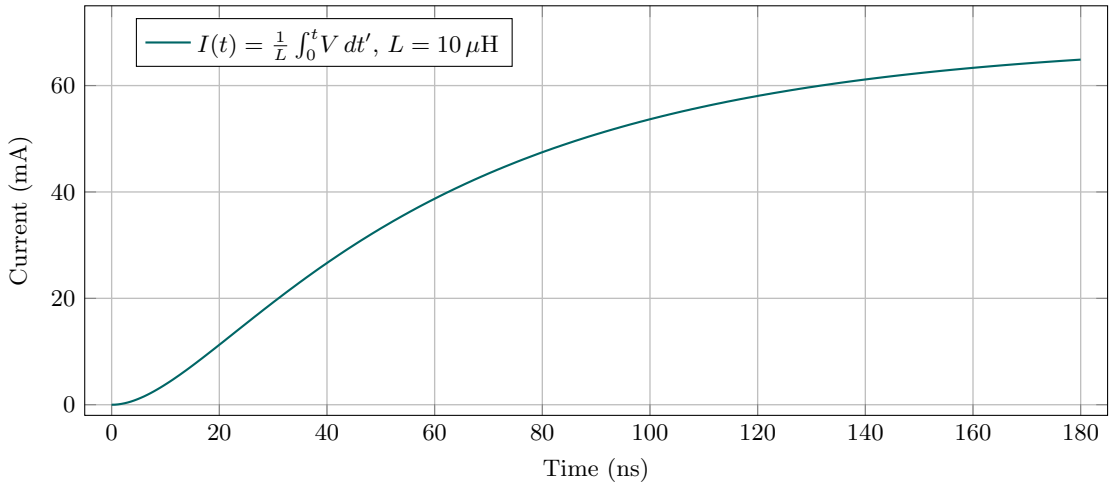


Figure 12: Current Estimate

Notice how the slope in the beginning is zero, this is because inductors strongly oppose changes in current, stopping initial change and creating a voltage spike. Also notice how the function resembles the how voltage changed in the capacitor. Simply put capacitors and inductors act the same under their properties they each depend on.

Now to use this to derive our circuits τ, we can start by relating the setting the equation for $I_L(t)$ equal to $\Delta_{\%}I_L(\infty)$ the currents on either side will cancel, letting us make an equation for tau in terms of the time and percentage of maximum current at the time.

$$\Delta_{\%}I_L(\infty) = I_L(\infty)(1 - e^{-\frac{t}{\tau}})$$
$$\Delta_{\%} = (1 - e^{-\frac{t}{\tau}})$$

$$1 - \Delta_{\%} = e^{-\frac{t}{\tau}}$$
$$\tau = \frac{-t}{ln(1 - \Delta_{\%})}$$

Our estimate function approaches $68.594mA$ at $I_L(\infty)$. I will pick three points on the curve and use our new equation for tau at all of them, I will average the results and call that our time constant estimate. The data for this is in Table 4.

	$t(ns)$	$I(t)mA$	$\Delta_{\%}$	τ estimate (ns)
1	40	26.653	38.855%	81.312
2	60	38.733	56.467%	72.145
3	80	47.466	69.198%	67.934

Average τ	73.797

Table 4: Tau Estimates

Now to compare with the true time constant, which is $\tau = \frac{L}{R}$. This becomes $66.667ns$ which is about 10% lower that the found time constant.

3.3 Conclusion

Unfortunately the time constant we found is very off from the real one. I think the main issue is due to residual error between our data points for voltage and the best fit equation that was created. in the early part of the equation the data points are multiple volts off. In the sections we used to derive the time constant, the values are much closer but still off due to the errors earlier on in the graph. If you look at the peak values the error is very noticeable, and there should of been a much higher voltage spike.

Other than that the similarities we found between inductors and capacitors are very interesting. They use similar equations and act identically with respect to their side of the duality. Because of the similarities, since inductor voltage is proportional to the derivative of its current, we could infer that a capacitors current is proportional to the derivative of its voltage if we didn't already know that for a fact. We can also use their similarities to assume with certainty that the current in a capacitor will make a similar spike like the voltage in the inductor.